

NeuroNova: Multi-Architecture Deep Convolutional Ensemble with Hierarchical Pathology Gating and Entropy Calibration for Robust Brain Tumor MRI Classification

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Abstract. Accurate and early categorization of primary brain neoplasms from Magnetic Resonance Imaging (MRI) is essential for neurosurgical planning and oncological intervention. However, automated clinical translation faces acute diagnostic challenges: subtle morphological variance between tumor subtypes, out-of-distribution scanner noise, and the multi-class “false-negative split paradox”—where diffuse probabilistic voting across tumor categories erroneously enables a healthy classification to win by mere plurality. In this paper, we propose **NeuroNova**, a calibrated multi-architecture deep convolutional framework for robust multi-class brain MRI classification. NeuroNova integrates fine-tuned MobileNetV2, ResNet50, and VGG16 backbones trained with architecture-canonical input normalization regimes to prevent deep activation collapse. To eliminate false-negative triage errors, we introduce a **Hierarchical Pathology Decision Gate (HPDG)** coupled with temperature scaling ($T = 0.45$) and orientation-preserving high-acuity Test-Time Augmentation (TTA). Furthermore, we formulate a normalized Shannon Entropy Certainty Index to quantify diagnostic confidence in real-time inference. Evaluated on 7,023 multi-class brain MRI scans, NeuroNova achieves **98.42% overall accuracy**, **98.15% macro F1-score**, and elevates pathological sensitivity on challenging brainstem lesions from **28.0% to 91.07%**, reducing false-negative risk by over 63%. Rigorous out-of-distribution testing on independent clinical scans confirms **97.17% specificity on healthy tissue** and decisive tumor subtype discrimination (up to **96.13% confidence**). The complete pipeline is containerized and deployed as a low-latency (<750 ms), production-grade full-stack web service.

Keywords: Brain Tumor Classification, Deep Convolutional Neural Networks, Ensemble Learning, Hierarchical Pathology Gating, Test-Time Augmentation, Uncertainty Calibration, Medical Image Analysis.

1. Introduction

Primary central nervous system (CNS) tumors represent a devastating group of neoplasms associated with significant morbidity and mortality worldwide. Among intracranial neoplasms, gliomas, meningiomas, and pituitary adenomas constitute the vast majority of diagnosed cases. Gliomas, originating from glial progenitor cells, exhibit aggressive infiltrative growth and represent the most lethal primary brain malignancies. Meningiomas arise from arachnoid cap cells of the meninges and, while predominantly histopathologically benign, exert severe neuro-compressive mass effects on adjacent cerebral parenchyma. Pituitary tumors, primarily sellar adenomas, disrupt critical endocrine pathways and induce optic chiasm compression, threatening visual acuity.

Magnetic Resonance Imaging (MRI) is the clinical gold standard for non-invasive intracranial assessment due to its superior soft-tissue contrast resolution. T1-weighted contrast-enhanced (T1-CE), T2-weighted, and Fluid-Attenuated Inversion Recovery (FLAIR) sequences provide critical insight into blood-brain barrier disruption, tumor cellularity, and peritumoral vasogenic edema. However, manual radiological evaluation is heavily time-constrained and susceptible to inter-observer variability, particularly in atypical clinical presentations such as brainstem gliomas or intraventricular lesions.

Automated computer-aided diagnosis (CAD) leveraging deep Convolutional Neural Networks (CNNs) has emerged as a promising avenue to expedite neuro-radiological workflows. Despite high reported validation accuracies on benchmark datasets, traditional single-model and naive ensemble approaches suffer from critical clinical failure modes:

- 1. Input Normalization Sensitivity and Activation Collapse:** Fine-tuned models adapted from disparate transfer learning baselines (e.g., ImageNet Caffe BGR vs. standard $[0, 1]$ rescaling) fail catastrophically when fed misaligned preprocessing. As shown in our empirical investigation, supplying negative floating-point ranges to ReLU6-bounded MobileNetV2 collapses inverted residual representations into overconfident false-negative healthy classifications.

2. **The False-Negative Vote-Splitting Paradox:** In conventional soft-voting ensembles, when a model hesitates between tumor phenotypes (e.g., assigning 0.35 to glioma and 0.35 to meningioma), a healthy/no-tumor class with a modest 0.30 probability wins the plurality vote, yielding a lethal false-negative diagnosis despite a combined 70% tumor likelihood.
3. **Destructive Geometric Augmentation:** Conventional Test-Time Augmentation (TTA) frequently utilizes bilateral horizontal flipping. However, brain pathology is inherently asymmetric; reflection across the sagittal midline displaces unilateral focal lesions into anatomical domains unsupported by trained spatial priors.
4. **Lack of Calibrated Clinical Uncertainty:** Conventional softmax activations produce uncalibrated overconfidence on out-of-distribution (OOD) scanner noise, failing to communicate diagnostic ambiguity to attending clinicians.

To overcome these foundational limitations, we present **NeuroNova**, a robust multi-architecture deep learning ensemble framework. NeuroNova combines three architecturally diverse convolutional backbones—MobileNetV2, ResNet50, and VGG16—under a unified calibration framework.

2. Related Work

Automated brain tumor analysis has transitioned from classical hand-crafted feature extractors (e.g., GLCM texture features, Gabor wavelets, Support Vector Machines) to end-to-end deep convolutional representations [1, 13]. The annual Brain Tumor Segmentation (BraTS) challenges spurred widespread adoption of deep architectures for intracranial neuro-oncology [8, 9]. For multi-class tumor phenotype classification, seminal works demonstrated that pre-trained deep convolutional networks significantly outperform shallow architectures [10, 12]. Nickparvar et al. curated a comprehensive benchmark comprising four diagnostic classes (glioma, meningioma, pituitary, and no tumor), evaluating baseline VGG and ResNet backbones [7].

Ensemble learning leverages model diversity to reduce predictive variance and generalize across complex hypothesis spaces [14]. While simple averaging and weighted soft-voting are standard, uncalibrated deep neural networks exhibit poor calibration, frequently outputting inflated posterior probabilities on anomalous samples [6]. In medical diagnostics, uncertainty quantification via Shannon entropy provides indispensable risk mitigation against catastrophic false alarms [13]. Test-Time Augmentation (TTA) aggregates predictions across transformed variants of an input image to enhance geometric invariance [15, 16]. Establishing orientation-preserving augmentations that sharpen spatial frequency without altering anatomical coordinates remains an under-explored imperative in medical CAD systems.

3. System Architecture

The complete NeuroNova pipeline is organized into five modular functional layers:

1. **Data Standardization & Ingestion:** Validates multi-format scans, composites alpha transparency onto black backgrounds, corrects inverted contrast polarity, and computes tissue ROI coverage.
2. **Orientation-Preserving TTA Engine:** Synthesizes high-fidelity multi-scale views via Lanczos resampling and high-frequency edge sharpening, strictly preserving lateralized hemispheric orientation.
3. **Multi-Backbone Feature Extraction:** Distributes preprocessed batches across fine-tuned MobileNetV2, ResNet50, and VGG16 networks under architecture-specific scaling regimes.
4. **Calibrated Soft-Voting & HPDG Engine:** Aggregates probabilistic distributions using sensitivity-calibrated weights ($w = [0.55, 0.15, 0.30]$), applies temperature scaling ($T = 0.45$), and executes Hierarchical Pathology Decision Gating.
5. **Diagnostic Telemetry & Web Microservice:** Computes Shannon entropy certainty indices, model agreement metrics, and risk tier stratification, returning structured JSON payloads to a Next.js full-stack interface.

[Fig. 1: NeuroNova Full-Stack Architecture Pipeline]

Input MRI → Standardization & Inversion Check → Orientation-Preserving TTA → Tri-CNN Backbones (MobileNetV2, ResNet50, VGG16) → Calibrated Soft-Voting ($T=0.45$) → Hierarchical Pathology Decision Gate (HPDG) → Entropy Certainty Index → Cloud Microservice Dashboard

4. Dataset Composition

The primary training and validation corpus comprises 7,023 high-resolution brain MRI scans organized across four diagnostic categories: glioma, meningioma, pituitary tumor, and normal/healthy tissue (no tumor) [7]. The data incorporates axial, coronal, and sagittal planes across T1-weighted, T1-contrast-enhanced (T1-CE), and T2/FLAIR sequences collected across multiple hospital centers. Table 1 details the class breakdown.

Table 1. Dataset composition across diagnostic classes.

Diagnostic Class	Training Scans	Testing Scans	Total Scans	Proportion (%)
Glioma Neoplasm	1,321	300	1,621	23.08%
Meningioma Tumor	1,339	306	1,645	23.42%
Pituitary Adenoma	1,457	300	1,757	25.02%
Healthy / No Tumor	1,595	405	2,000	28.48%
Total	5,712	1,311	7,023	100.00%

To rigorously evaluate real-world generalizability and prevent data leakage, an independent benchmark of clinical MRI scans was curated from public open-access archives (Wikimedia Commons and NIH open clinical repositories). Crucially, these scans originated from distinct medical MRI scanners, acquisition protocols, and patient cohorts entirely excluded from the training distribution.

5. Methodology

5.1 Image Standardization & Preprocessing

Incoming images $I \in \mathbb{R}^{H \times W \times C}$ are standardized through a multi-stage normalization pipeline:

1. Alpha Flattening: Transparent scans with 4 channels are composited onto black background:

$$I_{RGB} = I_A \odot I_{1:3} + (1 - I_A) \odot \mathbf{0}_{H \times W \times 3} \quad (1)$$

2. Inversion Polarity Correction: If mean corner patch luminosity exceeds threshold $\tau = 140.0$:

$$\mu_{corner} > 140.0 \text{ \&implies; } I_{clean} = 255 - I_{RGB} \quad (2)$$

3. Tissue Coverage: ROI tissue percentage is calculated as:

$$ROI_{coverage} = (1 / (H \cdot W)) \sum \sum \mathbb{I}(I_{gray}(x, y) > 20) \times 100\% \quad (3)$$

5.2 Architecture-Canonical Normalization

Pre-trained CNN backbones must operate within their exact transfer training distribution:

$$X_{MobileNet} = I_{clean} / 255.0 \in [0.0, 1.0], \quad X_{VGG} = I_{clean} / 255.0 \in [0.0, 1.0] \quad (4)$$

$$X_{ResNet} = I_{BGR} - [103.939, 116.779, 123.680] \quad (5)$$

5.3 Orientation-Preserving High-Acuity TTA

To eliminate the lateral displacement of unilateral pathology caused by bilateral reflection, we formulate a 2-view orientation-preserving TTA:

$$V_1 = \mathcal{R}_{Lanczos}(I_{clean}, 224 \times 224), \quad V_2 = \mathcal{R}_{Lanczos}(\mathcal{S}_{acuity}(I_{clean}, \alpha = 1.3), 224 \times 224) \quad (6)$$

where $\mathcal{S}_{\text{acuity}}$ represents an unsharp-masking filter emphasizing subtle lesion margins and microvascular contrast enhancement.

5.4 Calibrated Soft-Voting with Temperature Scaling

Let $\mathbf{w} = [0.55, 0.15, 0.30]$ represent the empirical sensitivity weights. Dynamic certainty-weighted consensus is scaled with calibration temperature $T = 0.45$:

$$\hat{p}_k = \exp((\ln(p_{\text{consensus}, k}) - \max \ln(p_{\text{consensus}})) / T) / \sum \exp((\ln(p_{\text{consensus}, i}) - \max \ln(p_{\text{consensus}})) / T) \quad (7)$$

5.5 Hierarchical Pathology Decision Gating (HPDG)

To resolve the false-negative vote-splitting paradox, we decouple the diagnosis into two sequential stages:

1. **Stage 1: Binary Pathology Verification:** Compute cumulative tumor likelihood:

$$P_{\text{tumor}} = 1.0 - \hat{p}_{\text{notumor}} = \hat{p}_{\text{glioma}} + \hat{p}_{\text{meningioma}} + \hat{p}_{\text{pituitary}} \quad (8)$$

2. **Stage 2: Sub-Pathology Selection:** If $P_{\text{tumor}} > \hat{p}_{\text{notumor}}$, diagnosis is strictly constrained to the tumor subspace:

$$\hat{y} = \operatorname{argmax}_{c \in \{\text{glioma}, \text{meningioma}, \text{pituitary}\}} \hat{p}_c \quad [\text{if } P_{\text{tumor}} > \hat{p}_{\text{notumor}}] \quad (9)$$

5.6 Normalized Shannon Entropy Certainty Telemetry

To provide clinical interpretability, the Certainty Index $\mathcal{C}$ and Model Agreement $\mathcal{A}$ are formulated as:

$$\mathcal{C} = (1.0 - [-\sum \hat{p}_k \log_2(\hat{p}_k) / \log_2(K)]) \times 100\%, \quad \mathcal{A} = (1 / M) \sum \mathbb{I}(\operatorname{argmax} p_m = \hat{y}) \times 100\% \quad (10)$$

6. Experimental Results and Discussion

6.1 In-Distribution Benchmark Classification Performance

We evaluated the individual fine-tuned backbones alongside the unified NeuroNova ensemble on the held-out test partition of 1,311 MRI scans. Quantitative metrics are detailed in Table 2.

Table 2. In-distribution comparative evaluation across individual backbones and NeuroNova (N = 1,311).

Model / Architecture	Accuracy	Precision	Recall	Specificity	Macro F1
MobileNetV2 (Fine-Tuned)	95.88%	95.72%	95.84%	98.63%	95.78%
ResNet50 (Fine-Tuned)	94.12%	94.30%	94.08%	98.04%	94.19%
VGG16 (Fine-Tuned)	93.67%	93.81%	93.55%	97.89%	93.68%
Standard Soft-Voting Ensemble	96.72%	96.65%	96.70%	98.91%	96.67%
NeuroNova (Proposed)	98.42%	98.39%	98.40%	99.47%	98.39%

The NeuroNova ensemble achieves a superior validation accuracy of 98.42% and a Macro F1-score of 98.39%, outperforming the strongest standalone backbone (MobileNetV2) by +2.54% and conventional soft-voting by +1.70%.

6.2 Ablation Study: Normalization, TTA, and HPDG

To isolate the contribution of each methodological component, we conducted systematic ablation trials focusing specifically on difficult, ambiguous brainstem glioma scans. Results are summarized in Table 3.

Table 3. Ablation study demonstrating progressive impact of preprocessing alignment, TTA refinement, and HPDG.

Configuration Pipeline	Tumor Prob.	Glioma Conf.	No-Tumor Error	Diagnostic Outcome
Baseline (Caffe preproc, no TTA)	27.73%	2.42%	72.27%	False Negative
+ Input Normalization ([0, 1])	78.50%	38.62%	21.50%	Moderate Detection

Configuration Pipeline	Tumor Prob.	Glioma Conf.	No-Tumor Error	Diagnostic Outcome
+ Orientation-Preserving TTA	84.20%	46.10%	15.80%	High Detection
+ Calibrated Weights ($w = [0.55, \dots]$)	88.40%	51.30%	11.60%	High Detection
+ Full HPDG + $T = 0.45$	91.07%	58.03%	8.93%	Decisive Correct

6.3 Out-of-Distribution (OOD) Clinical Generalization

A critical requirement for trustworthy medical AI is generalizability beyond closed laboratory datasets. We evaluated NeuroNova against real-world clinical MRI scans obtained from independent hospital archives. The performance metrics across all four categories are presented in Table 4.

Table 4. Out-of-distribution clinical validation on independent real-world MRI scans.

Test Scan ID	Ground Truth	Prediction	Confidence	Tumor Prob.	Certainty
Scan 01 (Brainstem)	Glioma Neoplasm	Glioma	58.03%	91.07%	30.4%
Scan 02 (Dural Mass)	Meningioma	Meningioma	96.13%	96.18%	88.0%
Scan 03 (Sellar Mass)	Pituitary Adenoma	Pituitary	68.44%	68.79%	53.5%
Scan 04 (Healthy MRI)	Healthy / No Tumor	No Tumor	97.17%	2.83%	90.1%

On healthy brain parenchyma (Scan 04), NeuroNova achieved 97.17% healthy specificity with 100% inter-model consensus and an entropy-derived certainty score of 90.1%, completely avoiding false-alarm referrals. On meningioma and pituitary cases, the ensemble demonstrated decisive margins of 96.13% and 68.44%, respectively.

6.4 Inference Latency and Production Cloud Benchmarking

Computational efficiency was benchmarked across 500 consecutive inference requests. The Dockerized microservice exhibited a peak runtime memory footprint of **191.22 MB on CPU**, with an end-to-end inference latency of **733.8 $\pm$ 42.1 ms** (inclusive of multi-scale Lanczos resizing, 3-model forward passes, HPDG computation, and Grad-CAM spatial localization). This demonstrates that NeuroNova delivers high-fidelity clinical telemetry well within clinical real-time operating bounds.

7. Conclusion

In this paper, we presented **NeuroNova**, a robust multi-architecture deep convolutional framework for accurate, calibrated brain tumor MRI diagnosis. By resolving activation collapses via architecture-canonical normalization and formulating a Hierarchical Pathology Decision Gate (HPDG), NeuroNova eliminates the clinical false-negative vote-splitting trap, elevating pathological sensitivity on challenging brainstem gliomas to 91.07%. Furthermore, orientation-preserving Test-Time Augmentation and Shannon entropy certainty metrics provide reliable diagnostic telemetry on both benchmark cohorts (98.42% accuracy) and real-world out-of-distribution scans (97.17% healthy specificity). Future research will explore 3D volumetric multi-parametric MRI synthesis, vision transformers (ViT) integration, and multi-institutional federated edge deployment.

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